

Puspak Chakraborty

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EDUCATION

Indian Institute of Technology Madras (IIT Madras)

Chennai, India

M.S. in Data Science and Artificial Intelligence

June 2024 – Present

CGPA: 9.06/10

RCC Institute of Information Technology

Kolkata, India

B.Tech in Computer Science

June 2016 – June 2020

CGPA: 9.33/10

RESEARCH EXPERIENCE

Indian Institute of Technology Madras

Chennai, India

M.S. Research Scholar

June 2024 – Present

- Investigating the theoretical and empirical equivalence between Deep Neural Networks (DNNs) and boosting algorithms.
- Developing Boosting architectures that use neural routers for intelligent spatial partitioning to handle heteroscedastic noise.
- **Publications:** One manuscript detailing the boosting architecture for heteroscedastic noise is currently under review.

WORK EXPERIENCE

Indian Institute of Technology Madras

Chennai, India

Teaching Assistant

January 2025 – present

- **Foundations of Machine Learning (DA5400):** Facilitated course operations and provided academic support for approximately 150 students (July 2025 – Nov 2025).
- **Mathematical Foundations of Data Science (DA5000):** Assisted in course operations for about 90 students (Jan 2025 – May 2025).

Infosys

Kolkata, India

Specialist Programmer

October 2020 – July 2024

- Developed serverless data pipelines using AWS SQS-triggered Lambda functions for a legal domain product.
- Designed REST APIs and optimized existing enterprise codebase, significantly improving processing efficiency.

PROJECTS

MeetGenAI: Advanced Meeting Intelligence Pipeline | *FastAPI, Whisper, Pyannote, ChromaDB, PyTorch* | github

- Engineered a parallelized audio processing pipeline combining Whisper for transcription, Pyannote for speaker diarization.
- Developed a multi-modal NLP module utilizing spaCy and SentenceTransformers to extract entities, track topic evolution, and perform sentiment analysis on time-chunked conversation segments.
- Implemented a Retrieval-Augmented Generation (RAG) architecture using ChromaDB to maintain historical meeting context, powering a fine-tuned Llama-3 model (via PEFT/LoRA) to generate structured executive summaries.

Handdrawn Image Classifier System with Feedback | *FastAPI, MLflow, Docker, Prometheus, DVC* | github

- Architected a containerized FastAPI web app for real-time doodle classification with human-in-the-loop feedback.
- Built an automated retraining pipeline using DVC, Git, and MLflow for seamless dataset versioning and model updates.
- Implemented system observability with Prometheus, Grafana, and a PostgreSQL-backed dashboard for pipeline tracking.

Seq2Seq Neural Machine Transliteration | *PyTorch, WandB, RNN/LSTM/GRU, Attention* | github | WandB

- Developed a Encoder-Decoder model (RNN/GRU/LSTM with Attention) for English to Hindi transliteration.
- Implemented greedy and beam search decoding strategies with length penalties.
- Orchestrated Bayesian hyperparameter tuning via Weights & Biases and built custom attention heatmap visualizations.

SKILLS

Languages: C++, Python, SQL

Tools & Frameworks: PyTorch, AWS, Git, MySQL, Prometheus, Grafana, Airflow, DVC

Core Concepts: Machine Learning, Artificial Intelligence, Optimization Algorithms

ACHIEVEMENTS & CERTIFICATIONS

- Secured Rank 11 out of 61 teams in KLA Image Restoration Hackathon, hosted by Computer Science Department, IIT Madras.
- Secured **All India Rank (AIR) 439**/123,967 candidates in **GATE 2024** (Computer Science), **GATE Score:** 765/1000.
- AWS Certified Cloud Practitioner (01/2022 – 01/2025).